2020 International Symposium on Nonlinear Theory and Its Applications,
NOLTA2020, Virtual, November 16-19, 2020

Application of Recurrence Quantification Analysis to Hypertension Photoplethysmograms

Nina Sviridova and Tohru Ikeguchi

Faculty of Engineering, Tokyo University of Science
6-3-1 Niijuku, Katsushika, Tokyo 125-8585, Japan
Email: nina.svr@rs.tus.ac.jp, tohru@rs.tus.ac.jp

Abstract– Hypertension is one of the widespread cardiovascular system disorders, which may cause various health complications. Hypertension can be diagnosed by the blood pressure (BP) measurement; however, conventional cuff-based devices cannot be used for continuous health monitoring. In recent studies, photoplethysmogram (PPG) was applied for cuff-less BP estimation. This PPG-based technique has great potential for application in the wearable devices for continuous real-time health monitoring and can be used for early identification of hypertension condition. Unlike the case of the clinical settings, real-time health monitoring by the PPG recorded by the wearable devices may require operating with the short PPG recordings. The PPG dynamics is highly complex, and the methods of nonlinear time series analysis can be efficiently used for its analysis. This data-driven study aimed to investigate the possibility of a recurrence quantification analysis application for the detection of hypertension conditions in short-recorded PPG signals. The results demonstrated that the mean diagonal line length, maximal diagonal line length, and the trapping time calculated from the recurrence phot can be potentially useful for differentiation between normal, prehypertension, stage 1 and 2 hypertension cases.

1. Introduction

Human photoplethysmogram (PPG) is one of the noninvasively measured biological signals widely applied to health monitoring. It is well known that PPG carries rich information on the performance of the cardiovascular system [1]. The heart rates and oxygen estimation have been main PPG applications for decades, and recent studies demonstrated that PPG can also be used for cuff-less blood pressure (BP) estimation [2]. High blood pressure or hypertension is the globally spread cardiovascular system disorder, which depends on its category may be potentially harmful to health, as well as lead to or worsen other cardiovascular system diseases. Timely identification of hypertension conditions is an important issue. Continuous real-time health monitoring by wearable devices, such as smartwatches, could be its possible solution. However, the number of issues naturally arises when the PPG measured by the wearable device is used. One of them is that unlike the PPG measured in clinical settings, where the measurement protocol and devices are unified, the wearable devices, that are the most promising in terms of health monitoring, have a wide range of device setups and the types of the PPG used. To address this issue one may need to rely on the essential characteristics of the PPG signal dynamics. The most commonly used NIR and green light PPGs are known to have chaotic dynamics [3-5]; therefore many studies have been conducted towards the identification of various physiological and mental health disorders by the nonlinear time series analysis of the PPG signals [3, 6, 7], that can reveal its underlying dynamical characteristics. Recurrence quantification analysis (RQA) [8] is one of these methods [9].

Another important issue is that real-time health monitoring by signals continuously recorded during daily activities may require to operate with short recordings, as the PPG signal may be corrupted by the motion artifacts, presence of measurement noise, etc.

Taking the above-mentioned issues into consideration, this data-driven study aims to investigate whether the RQA is effective for the identification of the differences in the dynamics of the short recorded PPGs obtained from hypertension condition subjects in comparison with healthy subjects' PPGs.

2. Data

In this study, the short-recorded photoplethysmogram database for blood pressure monitoring [10] was utilized. Each dataset was recorded for 2.1s with sampling frequency 1 kHz. Detailed experimental protocol, database description, its availability and data quality check information can be found in [11].

For this study data of subjects with normal, prehypertension (preH), stage 1 hypertension (S1H), stage 2 hypertension (S2H) records, and with no records of diabetes, cerebral infarction, and cerebrovascular disease were selected.

It is well-known that the PPG data may significantly change with age [12]. To reduce influence of the subjects' age, among the 219 subjects (age range: 21-86) datasets provided in the database, the data corresponding to the subjects over 45 years old were chosen for this study. Used data age range, the total number of datasets (N), average BP for each PPG group are summarized in table 1.

Table 1. Summary of hypertension groups records

Hypertension	N	Age	Average BP, mmHg
Normal	49	45-75	110/64
Prehypertension	73	45-84	130/72
Stage 1 hypertension	31	46-77	148/82
Stage 2 hypertension	20	45-80	168/89

A typical example of the raw data from normal and S2H group is shown in Fig. 1.

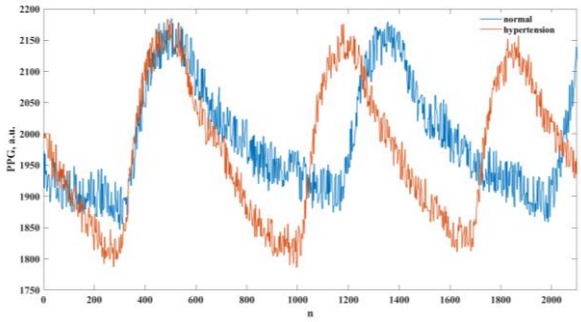


Fig. 1. A typical example of unprocessed PPG signal for normal (blue line) and stage 2 hypertension (red line) subject.

3. Results

3.1. Recurrence plot

The recurrence plot was calculated for each dataset. Based on the previous studies [4, 5], the applied embedding dimension was chosen equal to 4. Fig. 2 demonstrates an example of the obtained recurrence plots corresponding to the normal, preH, S1H, and S2H data.

3.2. Recurrence Quantification Analysis (RQA)

The RQA was applied to the calculated recurrence plots to get the quantitative characteristics of PPGs underlying dynamics. The RQA was conducted by the MATLAB CRP toolbox [8]. The following indexes were calculated: recurrence rate (RR), determinism (DET), mean diagonal line lengths (L), maximal diagonal line lengths (L_{max}), laminarity (LAM), trapping time (TT), and maximal vertical line lengths (V_{max}). The mean values of these indexes for each group are shown in table 2. Time series which DET values were below 0.9 were omitted from the study due to the possibility of significant noise contamination.

First, a one-factor ANOVA test was applied to each index to test the null hypothesis of equal means for four hypertension groups. The null hypothesis was confirmed for the RR and DET indexes; therefore, the difference in the mean values of these indexes is not significant; for other indexes the null hypothesis was rejected.

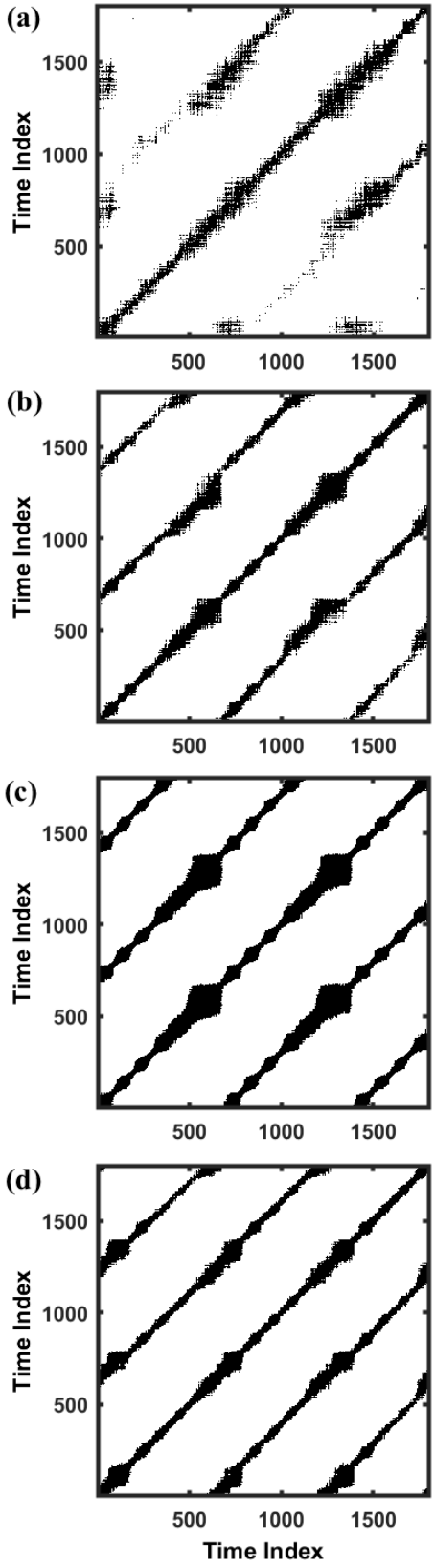


Fig. 2. An example of recurrence plot for the PPG signal corresponding to the (a) normal, (b) prehypertension, (c) stage 1 hypertension and (b) stage 2 hypertension subject.

Table 2. Mean values of calculated RQA indexes

	Normal	PreH	S1H	S2H
RR	0.15	0.15	0.15	0.16
DET	0.98	0.98	0.98	0.99
L	27.44	25.67	25.09	28.46
L_{max}	1600.78	1684.16	1742.39	1799.00
ENTR	2.98	2.94	2.93	3.03
LAM	0.99	0.99	0.99	1.00
TT	32.28	31.19	30.72	34.05
V_{max}	32.28	31.19	30.72	34.05

T-test was applied to the rest of the RQA indexes to test the null hypothesis on the equal means between pairs of different hypertension groups. The results demonstrated that none of the applied indexes could differentiate between S1H and S2H groups. For the following indexes and pairs of groups the null hypothesis was rejected with significance level 0.05:

- normal vs. preH: L, L_{max}, ENTR, LAM, TT, V_{max},
- normal vs. S1H: L, L_{max}, TT,
- normal vs. S2H: L, L_{max}, ENTR, LAM, TT,
- preH vs. S1H: L, L_{max}, ENTR, LAM, TT,
- preH vs. S2H: L, L_{max}, ENTR, LAM, TT.

4. Discussion

The aim of this study was to investigate whether the RQA indexes can be applied to differentiate the short-recorded PPG data obtained from subjects with normal, prehypertension, stage 1 and stage 2 hypertension. As seen from the typical examples of time series (Fig. 1) and different hypertension groups recurrence plots (Fig. 2), no significant qualitative difference can be observed. However, RQA results demonstrated that significant differences can be found in characteristics of the PPGs corresponding to normal, prehypertension, and hypertension data. The L, Lmax, ENTR, LAM, and TT RQA indexes were found to be applicable to identify differences in the dynamics of studied data. However, no statistical differences were observed for the RR and DET indexes; and none of the applied indexes was able to differentiate between stage 1 and 2 hypertension groups.

It worths noticing that most of the PPG datasets for all groups are deterministic as follows from the results of DET calculation.

5. Conclusion

Results of this data-driven study demonstrated that the RQA measures, namely, mean diagonal line lengths, maximal diagonal line length, and the trapping time can potentially be applied for identification of the PPGs corresponding to the subjects with hypertension disorder. This result can contribute to the area of real time health monitoring and early identification of hypertension.

Acknowledgments

This work was partially supported by JSPS KAKENHI Grant Number 19K14589 and 17K00348.

References

[1] Allen, J. "Photoplethysmography and its application in clinical physiological measurement," Physiological measurement, 2007; 28: 1-39.

[2] D. Fujita, A. Suzuki, K. Ryu, "PPG-Based Systolic Blood Pressure Estimation Method Using PLS and Level-Crossing Feature," *Appl. Sci.*, vol. 9, pp. 304 1-11, 2019.

[3] I. Tsuda, "Chaotic pulsation in human capillary vessels and its dependence on mental and physical conditions," *International journal of bifurcation and chaos*, vol. 2(2), pp. 313–324, 1992.

[4] N. Sviridova, K. Sakai, "Human photoplethysmogram: new insight into chaotic characteristics," *Chaos, solitons and fractals*, vol.77, pp. 53–63, 2015.

[5] N. Sviridova, T. Zhao, K. Aihara, A. Nakano, K. Nakamura, "Photoplethysmogram at green light: Where does chaos arise from?" *Chaos, solitons and fractals*, vol.116, pp. 157–165, 2018.

[6] T. D. Pham, T.C. Thang, M. Oyama-Higa, M. Sugiyama, "Mental-disorder detection using chaos and nonlinear dynamical analysis of photoplethysmographic signal," *Chaos, solitons & fractals*, vol. 51, pp. 64–74, 2013.

[7] N. Sviridova, K. Sakai, "Application of photoplethysmogram for detecting physiological effects of tractor noise," *Engineering in agriculture, environment and food*, vol. 8(4), pp. 313–317, 2015.

[8] N. Marwan, M. C. Romano, M. Thiel, J. Kurths, "Recurrence plots for analysis of complex systems," *Physics reports*, vol. 438, pp. 237–329, 2007.

[9] N. Sviridova, "Detection of Preprocessing-Induced Changes in Chaotic Characteristics of Human Photoplethysmogram". *Proceedings of Int'l Symposium on Nonlinear Theory and its Applications 2019*, pp. 540-543, 2019.

[10] Y. Liang, G. Liu, Z. Chen, M. Elgendi, PPG-BD database. Figshare. Dataset. https://doi.org/10.6084/m9.figshare.5459299.v3, 2018.

[11] Y. Liang, Z. Chen, G. Liu, M. Elgendi, "A new, short-recorded photoplethysmogram dataset for blood pressure monitoring in China," *Scientific data*, vol. 5, pp. 180020 1-7, 2018.

[12] J. Allen, A. Murray, "Age-related changes in peripheral pulse timing characteristics at the ears, fingers and toes," *J Hum Hypertens*, vol. 16, pp.11–17, 2002.